INTRODUCTION TO DATAWAREHOUSING

1. DECISION SUPPORT SYSTEM (DSS)

- Helps organizations make better decisions using data and models.
- Uses data, models, analytics, reports and dashboards.
- Types of decisions:
  - Strategic
  - Tactical
  - Operational

2. OPERATIONAL vs DECISION SUPPORT SYSTEM

OPERATIONAL SYSTEM (OLTP)

- Day-to-day operations
- Handles current data
- High volume transactions

Examples: Banking, Reservation, Order Entry

DECISION SUPPORT SYSTEM (DSS)

- Decision making & analysis
- Uses historical data
- Complex queries & analysis

Examples: Sales Analysis, Trend Forecasting

3. ARCHITECTURE OF DATA WAREHOUSE

Data Sources (Operational DB, Files, External Sources)

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ETL Process (Extract, Transform, Load)

↓

Data Warehouse (Cleaned, Integrated Data)

↓

OLAP Server (Processing & Analysis)

↓

Front-end Tools (Reports, Dashboards, Visualization)

4. ETL (EXTRACT, TRANSFORM, LOAD)

EXTRACT → TRANSFORM → LOAD

EXTRACT
- Data from multiple sources

TRANSFORM
- Clean
- Filter
- Integrate
- Aggregate

LOAD
- Into Data Warehouse

Purpose: Ensure data quality and consistency

5. OLAP AND DATA CUBES

OLAP (Online Analytical Processing)

- Multi-dimensional analysis of data
- Operations: Drill-down, Roll-up, Slice, Dice, Pivot

DATA CUBE

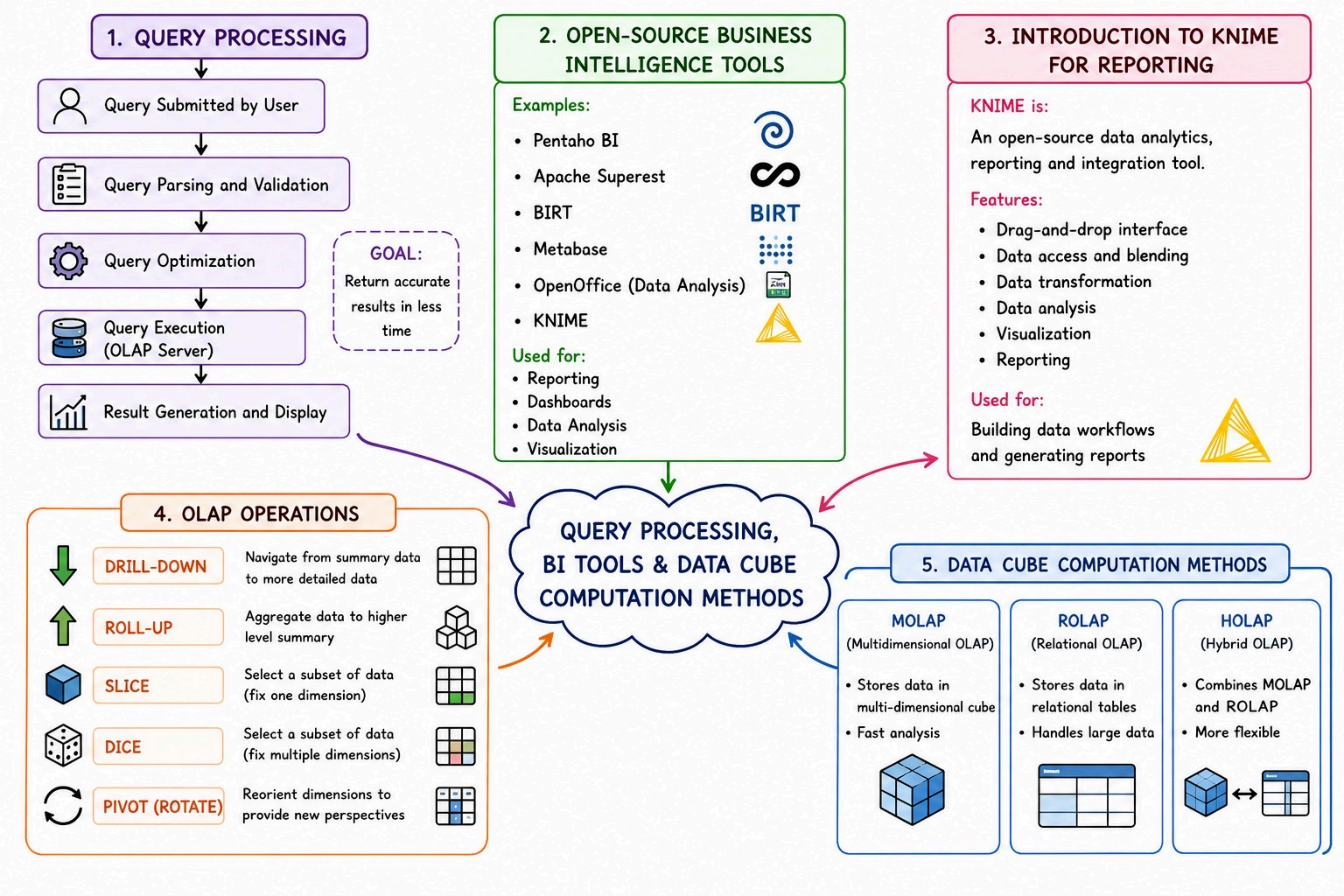
- Stores data in multi-dimensional structure
- Measures / Facts described by Dimensions

6. DIMENSIONAL DATA MODELLING

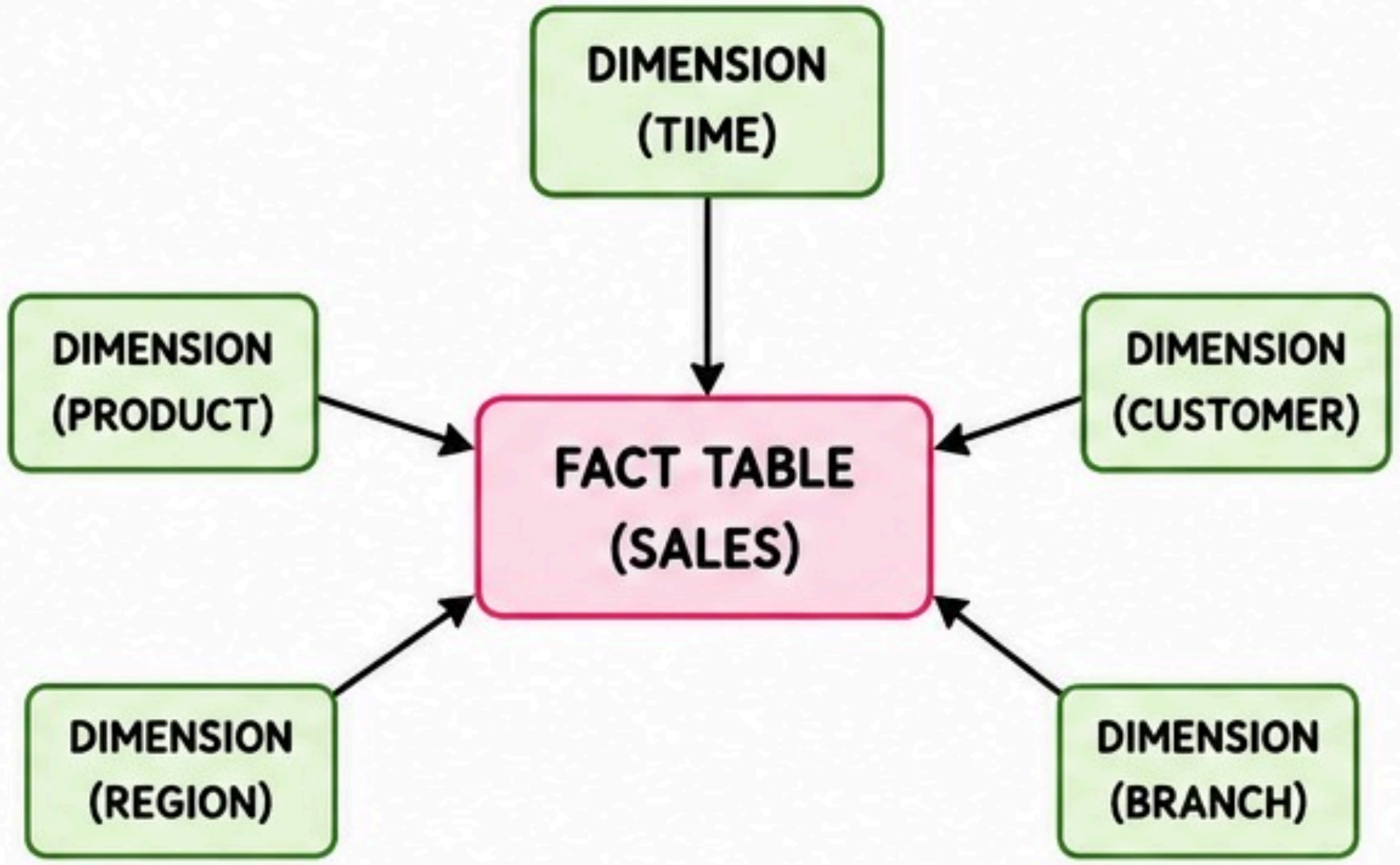
Used in Data Warehouse

- Star Schema
- Snowflake Schema

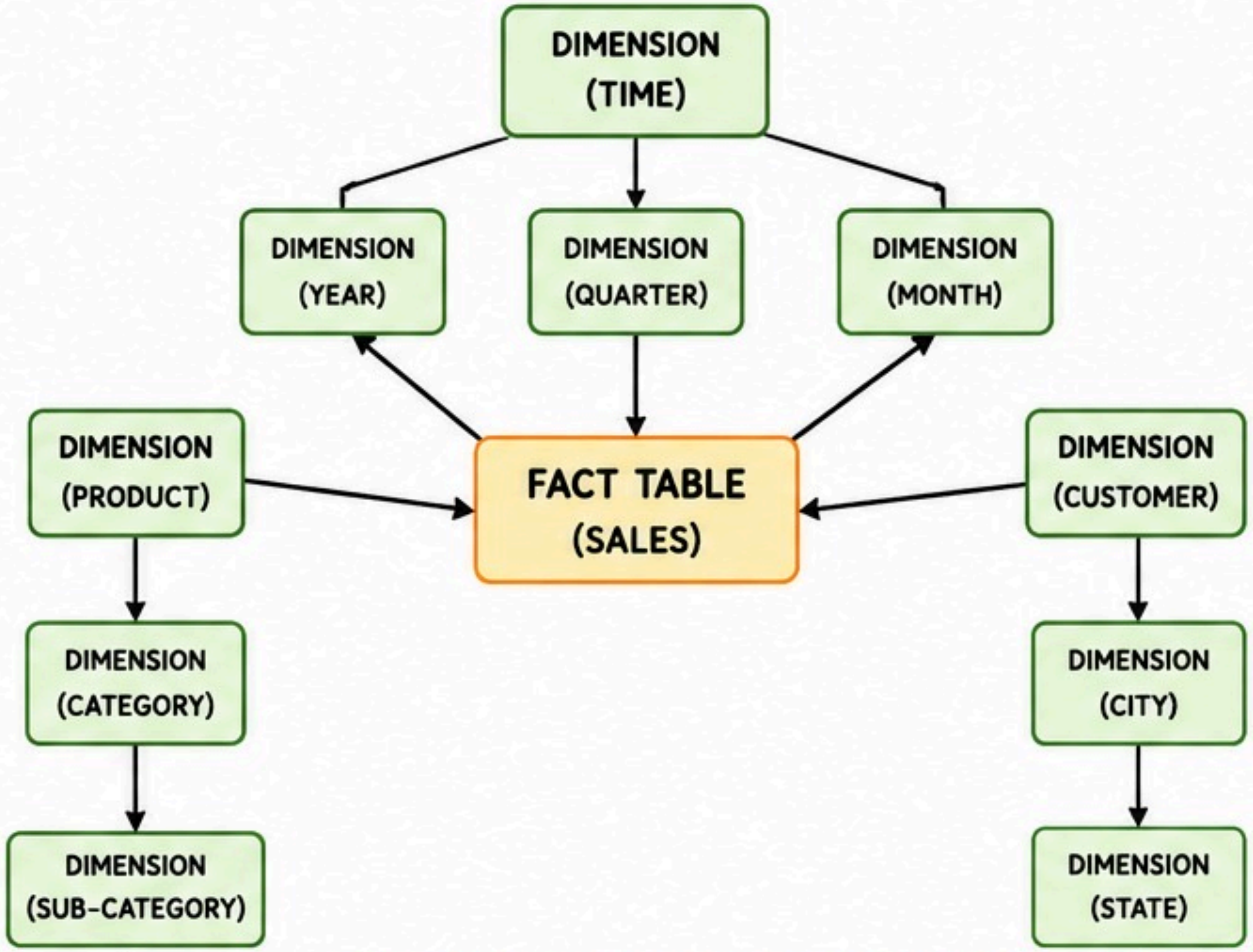
- Dimension tables are normalized into sub-dimensions


1. QUERY PROCESSING
Query Submitted by User
Query Parsing and Validation
Query Optimization
Query Execution (OLAP Server)
Result Generation and Display
GOAL:
Return accurate results in less time
2. OPEN-SOURCE BUSINESS INTELLIGENCE TOOLS
Examples:
• Pentaho BI
• Apache Superest
• BIRT
• Metabase
• OpenOffice (Data Analysis)
• KNIME
BIRT
Used for:
• Reporting
• Dashboards
• Data Analysis
• Visualization
3. INTRODUCTION TO KNIME FOR REPORTING
KNIME is:
An open-source data analytics, reporting and integration tool.
Features:
• Drag-and-drop interface
• Data access and blending
• Data transformation
• Data analysis
• Visualization
• Reporting
Used for:
Building data workflows and generating reports
QUERY PROCESSING, BI TOOLS & DATA CUBE COMPUTATION METHODS
4. OLAP OPERATIONS
DRILL-DOWN
Navigate from summary data to more detailed data
ROLL-UP
Aggregate data to higher level summary
SLICE
Select a subset of data (fix one dimension)
DICE
Select a subset of data (fix multiple dimensions)
PIVOT (ROTATE)
Reorient dimensions to provide new perspectives
5. DATA CUBE COMPUTATION METHODS
MOLAP
(Multidimensional OLAP)
• Stores data in multi-dimensional cube
• Fast analysis
ROLAP
(Relational OLAP)
• Stores data in relational tables
• Handles large data
HOLAP
(Hybrid OLAP)
• Combines MOLAP and ROLAP
• More flexible

1. STAR SCHEMA



2. SNOWFLAKE SCHEMA



- One fact table in the center connected to denormalized dimension tables.
- Simple and easy to understand.
- Faster query performance.
- Redundant data in dimension tables.

DIMENSIONAL DATA MODELLING

- Fact table connected to normalized dimension tables (sub-dimensions).
- Saves storage space.
- More complex than Star Schema.
- Slower queries compared to Star Schema.

FACT TABLE (CONTAINS MEASURES / FACTS)

Measures / Facts described by Dimensions
Example: Sales Amount, Quantity, Profit